MSG No. 135.490 $P4'_2/m'bc'$ [Type III, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `2/m'..`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,13,14]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,4,15,16]`
3	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[5,6,9,10]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[7,8,11,12]`

Table 2: Wyckoff site: `4b`, site symmetry: `-4..`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,5,6]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[3,4,7,8]`
3	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[9,10,13,14]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[11,12,15,16]`

Table 3: Wyckoff site: `4c`, site symmetry: `2/m'..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,2,13,14]`
2	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,4,15,16]`
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[5,6,9,10]`
4	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[7,8,11,12]`

Table 4: Wyckoff site: `4d`, site symmetry: `2.22`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,2,3,4]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[5,6,7,8]`
3	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[9,10,11,12]`
4	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[13,14,15,16]`

Table 5: Wyckoff site: `8e`, site symmetry: `2..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[3,4]`
3	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[5,6]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[7,8]`
5	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[9,10]`
6	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[11,12]`
7	$[0,\ 0,\ -z]$	`[13,14]`

8	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[15,16]

Table 6: Wyckoff site: 8f, site symmetry: 2..

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	[1,2]
2	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[3,4]
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	[5,6]
4	$\left[\frac{1}{2},\ 0,\ z\right]$	[7,8]
5	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	[9,10]
6	$\left[\frac{1}{2},\ 0,\ -z\right]$	[11,12]
7	$\left[0,\ \frac{1}{2},\ -z\right]$	[13,14]
8	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[15,16]

Table 7: Wyckoff site: 8g, site symmetry: ..2

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[1,3]
2	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[2,4]
3	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{4}\right]$	[5,8]
4	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{4}\right]$	[6,7]
5	$\left[\frac{1}{2}-x,\ x,\ \frac{3}{4}\right]$	[9,12]
6	$\left[x+\frac{1}{2},\ -x,\ \frac{3}{4}\right]$	[10,11]
7	$\left[-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[13,15]
8	$\left[x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[14,16]

Table 8: Wyckoff site: 8h, site symmetry: m'..

No.	position	mapping
1	$[x,\ y,\ 0]$	[1,14]
2	$[-x,\ -y,\ 0]$	[2,13]
3	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[3,16]
4	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[4,15]
5	$\left[y,\ -x,\ \frac{1}{2}\right]$	[5,10]
6	$\left[-y,\ x,\ \frac{1}{2}\right]$	[6,9]
7	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ 0\right]$	[7,12]
8	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	[8,11]

Table 9: Wyckoff site: 16i, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-x,\ -y,\ z]$	[2]
3	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[3]

continued ...

Table 9

No.	position	mapping
4	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[4]
5	$\left[y,\ -x,\ \frac{1}{2}-z\right]$	[5]
6	$\left[-y,\ x,\ \frac{1}{2}-z\right]$	[6]
7	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	[7]
8	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$	[8]
9	$\left[-y,\ x,\ z+\frac{1}{2}\right]$	[9]
10	$\left[y,\ -x,\ z+\frac{1}{2}\right]$	[10]
11	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[11]
12	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ -z\right]$	[12]
13	$[-x,\ -y,\ -z]$	[13]
14	$[x,\ y,\ -z]$	[14]
15	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[15]
16	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[16]